Fakulteta za informacijske študije
Faculty of information studies

lab06

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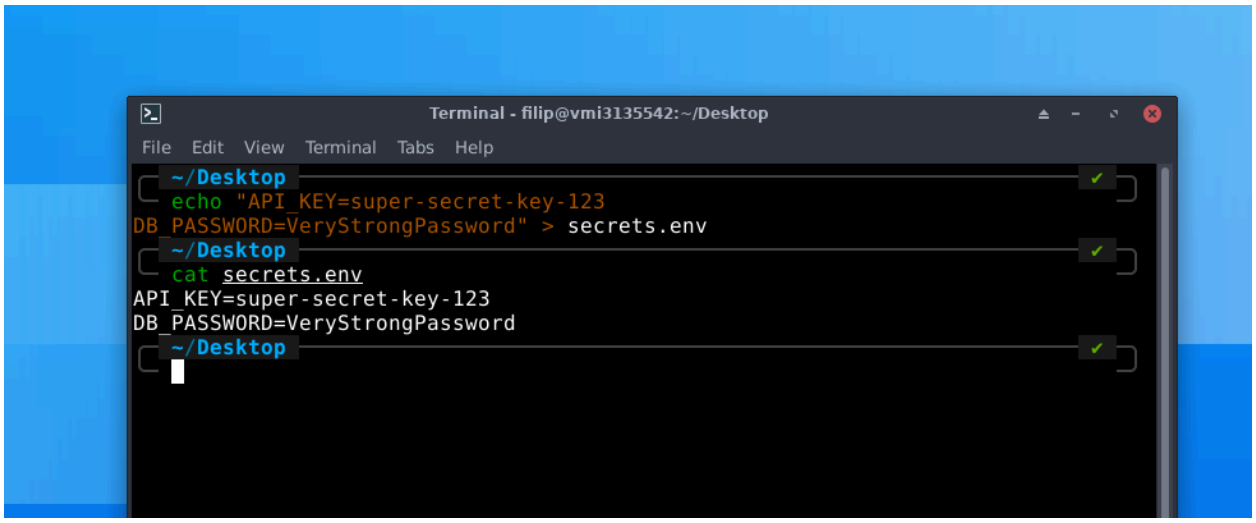

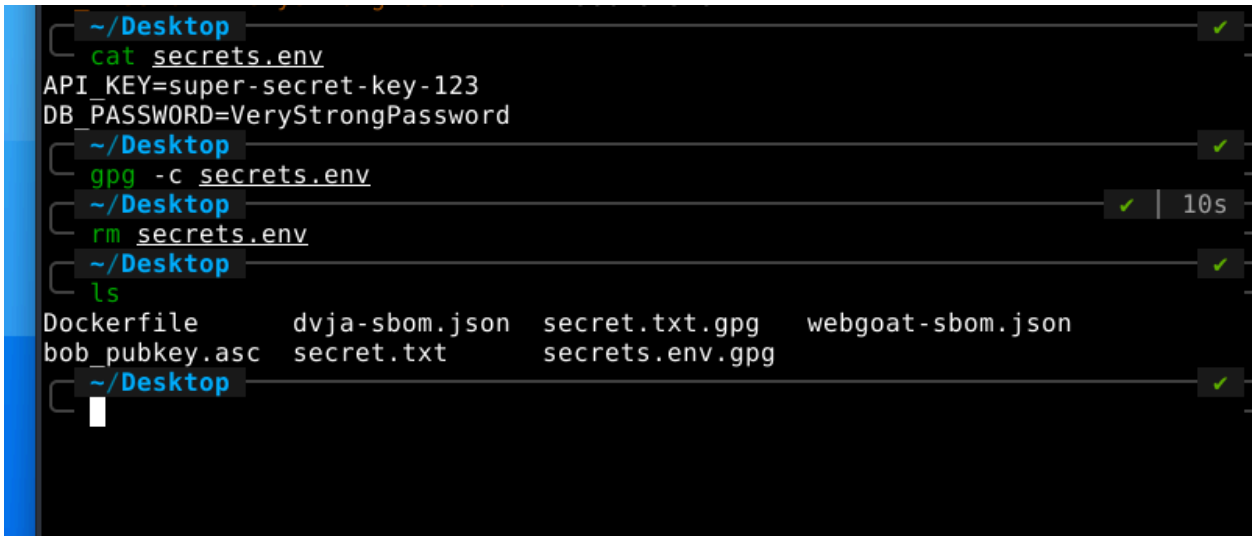

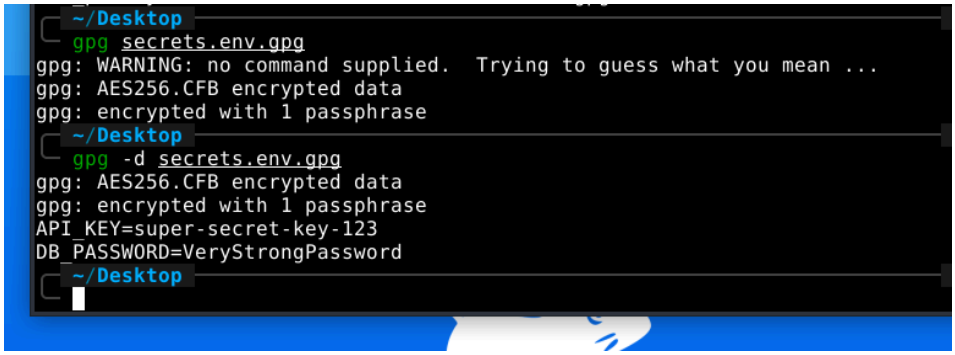

```
~/Desktop
echo "API_KEY=super-secret-key-123
DB_PASSWORD=VeryStrongPassword" > secrets.env
~/Desktop
gpg --encrypt --recipient alice@example.com secrets.env
```

```
~/Desktop
source secrets.env
echo $API_KEY
super-secret-key-123
~/Desktop
unset API_KEY
unset DB_PASSWORD
~/Desktop
```

```
~/Desktop                                                    ✔ | 12s
gpg secrets.env.gpg
gpg: WARNING: no command supplied.  Trying to guess what you mean ...
gpg: encrypted with rsa4096 key, ID 45B0DF4F07F1C2E9, created 2026-04-27
      "Alice <alice@example.com>"
File 'secrets.env' exists. Overwrite? (y/N)
```

```
File 'secrets.env' exists. Overwrite? (y/N) y
~/Desktop                                                    ✔ | 2m 54s
gpg --encrypt --recipient alice@example.com --recipient bob@example.com secre
ts.env
File 'secrets.env.gpg' exists. Overwrite? (y/N) y
~/Desktop                                                    ✔
gpg --decrypt secrets.env.gpg | source /dev/stdin
gpg: encrypted with rsa4096 key, ID 048FA5A2F69ED29E, created 2026-04-27
      "Bob <bob@example.com>"
gpg: encrypted with rsa4096 key, ID 45B0DF4F07F1C2E9, created 2026-04-27
      "Alice <alice@example.com>"
~/Desktop                                                    ✔
```

Why don't secrets belong in the source code?

Secrets do not belong in source code because repositories can be accessed by unauthorized users. If secrets are exposed, attackers can gain access to systems, APIs, or databases.

Difference between symmetric and asymmetric encryption?

Symmetric encryption uses a single password for both encryption and decryption. Asymmetric encryption uses a public key for encryption and a private key for decryption, which is more secure because the private key is never shared.

What happens if we lose the private key?

If the private key is lost, encrypted data cannot be decrypted. This means the secrets are permanently inaccessible.

How to handle in enterprise?

In larger enterprises, secrets are managed using specialized tools like HashiCorp Vault, AWS Secrets Manager, or SOPS. These tools provide secure storage, access control, auditing, and automated rotation of secrets.